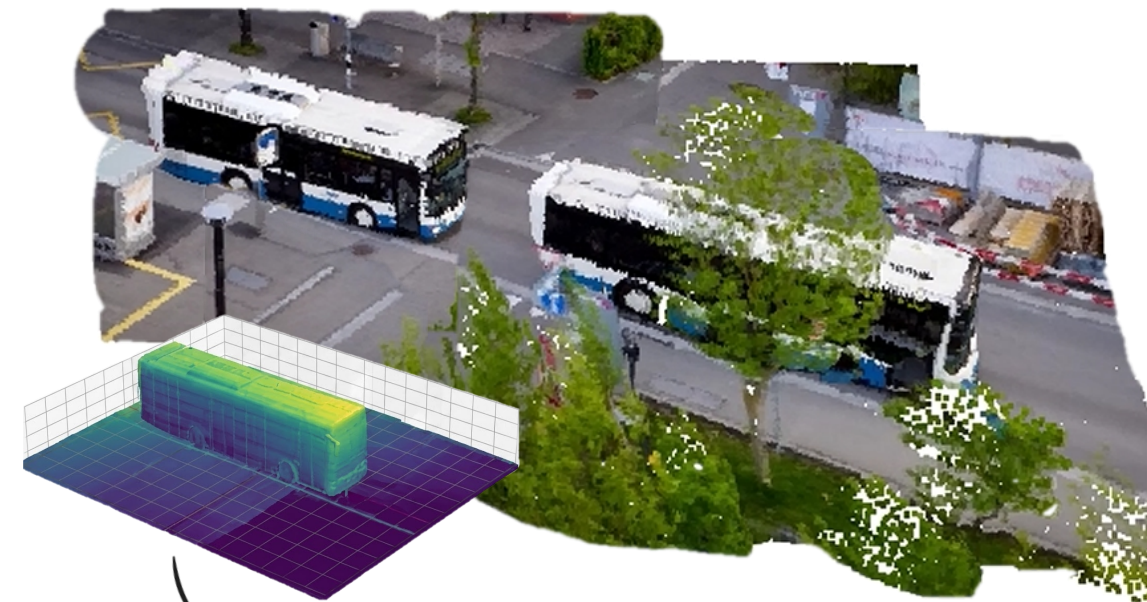


Step 0: Input Video



Step 2: 3D Geometric Uplifting (MegaSaM)



$\{Z_t\}_{t=1}^T, f$ & camera poses

World-space Pointmaps: $\mathbf{P}_{world} \in \mathbb{R}^{T \times H \times W \times 3}$

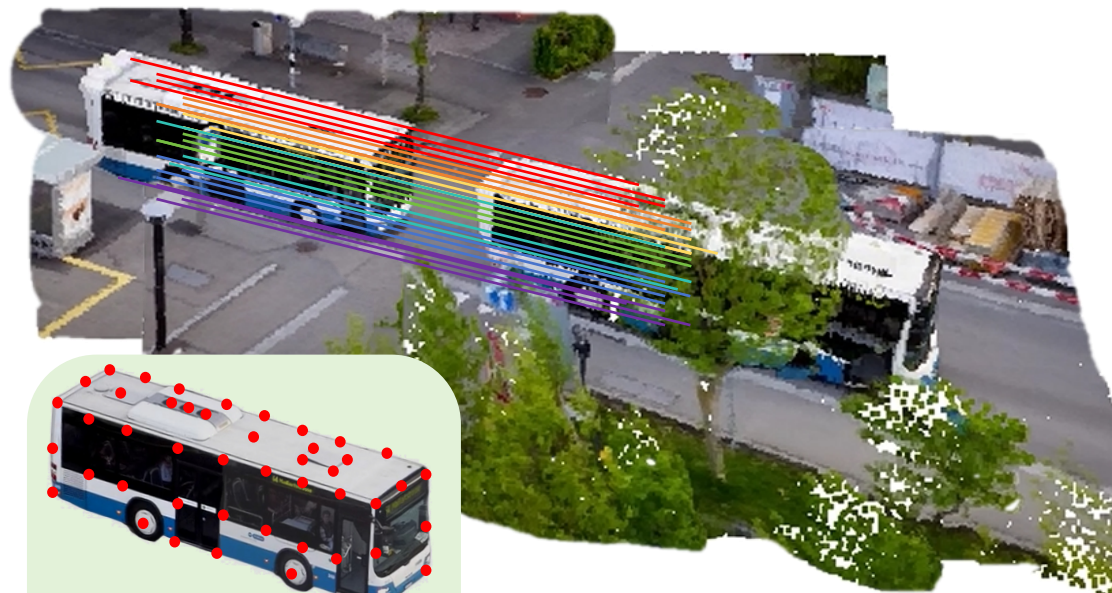
Step 1: Semantic Targeting (SAM 2)



prompts



Step 3: 3D Structural Anchoring (CoTracker3)



Anchor seeding
SIFT $\rightarrow$ Shi-Tomasi
 $\rightarrow$ uniform-grid

2D pixel-space trajectories
 $(u_t^n, v_t^n) \rightarrow \mathbf{q}_t^n = \mathbf{P}_{world}[t, v_t^n, u_t^n]$